OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 29 OF 55

AI-BASED SIGN LANGUAGE TRANSLATION WITH REAL-TIME SPEECH AND TEXT GENERATION

Domain: Computer Vision & Assistive Technology | **Module:** project_29_sign_language

OFFICIAL ABSTRACT

AI-Based Sign Language Translation with Real-Time Speech and Text Generation is a comprehensive and crucial project aimed at addressing communication barriers, accessibility gaps, and social inclusion challenges in modern communities and digital environments. As digital communication accelerates, societies increasingly rely on inclusive interactions, making the deaf and hard-of-hearing community prime targets for communication isolation if gestural translation is not proactively supported. Traditional translation methods often rely on manual interpretation or static phrasing, which can be time-consuming, resource-intensive, and fundamentally inefficient against rapidly moving, complex gestural movements and subtle sign variations. To address these persistent challenges and limitations, this study proposes an advanced, multi-layered approach combining cutting-edge deep learning techniques with intelligent machine learning algorithms for efficient, automated, and real-time sign language recognition and generation. The system utilizes specialized deep learning architectures to continuously capture and extract key behavioral features from extensive video streams and sensor inputs, such as fluctuating hand configurations, arm movements, facial expressions, and spatial trajectory deviations. Once these underlying structural patterns are mapped, the extracted features are fed into classification and translation algorithms designed to enhance decision-making, optimize computational overhead, improve prediction accuracy, and drastically reduce false alarm and misclassification rates. By profiling what constitutes normal, routine signs for each distinct vocabulary item, the model establishes a dynamic baseline that adapts to changing user styles over time. The proposed model is rigorously trained, validated, and tested on robust datasets encompassing diverse sign language video logs, operational user cycles, and simulated translation scenarios. This enables the framework to accurately distinguish between neutral gestures and critical communication tokens—such as specific words, phrases, and grammatical structures—with high precision and minimal latency. Experimental results

and comprehensive performance evaluations demonstrate that the seamless integration of AI-driven feature extraction and deep learning models provides significantly improved classification and regression performance, resilience, and adaptability compared to traditional legacy translation approaches.

📌 Project Summary & System Specifications

Project Title	AI-BASED SIGN LANGUAGE TRANSLATION WITH REAL-TIME SPEECH AND TEXT GENERATION
Specialization / Category	Computer Vision & Assistive Technology
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5029`
Local Implementation	`project_29_sign_language`